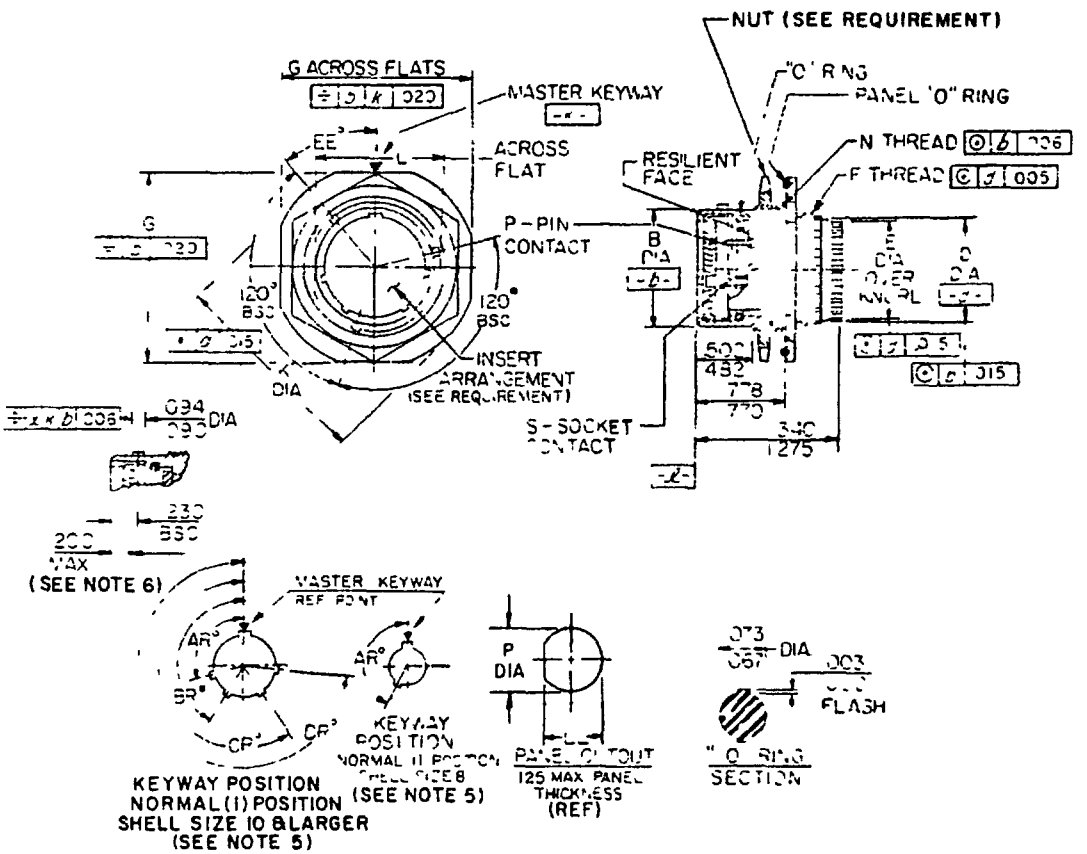
MIL-C-81511/3E
30 October 1970
SUPERSEDING
MIL-C-81511/3D
5 June 1970

MILITARY SPECIFICATION SHEET

CONNECTORS ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISCONNECT, ENVIRONMENT RESISTING, AND ACCESSORIES; RECEPTACLE, JAM NUT MOUNT, CRIMP-TYPE CONTACTS, CLASS A, F AND E (SERIES 2)

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-81511.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



FSC 5935

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TABLE I.

Shell size	B dia	D dia	E dia	F thread	G across flats	K dia	L across flat	LL	N thread	P dia
8	.553 (14.05) .547 (13.89)	.430 (10.92) .424 (10.77)	.422 (10.72) .407 (10.34)	½-28UNEF-2A	.989 (25.12) .969 (24.61)	1.072 (27.23) 1.052 (26.72)	.598 (15.19) .590 (14.99)	.610 (15.49) .600 (15.24)	5/8-20UN-2A	.640 (16.26) .630 (16.00)
10	.678 (17.22) .672 (17.07)	.555 (14.10) .549 (13.94)	.548 (13.92) 533 (13.54)	5/8-28UN-2A	1.114 (28.30) 1.094 (27.79)	1.197 (30.40) 1.177 (29.90)	.723 (18.36) .715 (18.16)	.735 (18.67) .725 (18.42)	3/4-20 UNEF-2A	.765 (19.43) .755 (19.18)
14	.928 (23.57) .922 (23.42)	.805 (20.45) .799 (20.29)	.798 (20.27) .783 (19.89)	7/8-28UN-2A	1.401 (35.59) 1.381 (35.08)	1.510 (38.35) 1.490 (37.85)	.973 (24.71) .965 (24.51)	.985 (25.02) .975 (24.77)	1–20 UNEF-2A	1.015 (25.78) 1.005 (25.53)
16	1.054 (26.77) 1.048 (26.62)	.930 (23.62) .924 (23.47)	.924 (23.47) .909 (23.09)	1–28UN-2A	1.576 (38.76) 1.506 (38.25)	1.635 (41.53) 1.615 (41.02)	1.098 (27.89) 1.090 (27.69)	1.110 (28 19) 1.100 (27.94)	1-1/8-20UN-2A	1.140 (28.96) 1.130 (28.70)
18	1.178 (29.92) 1.172 (29.77)	1.055 (26.80) 1.049 (26.64)	1.048 (26.62) 1 033 (26 24)	1-1/8-28UN-2A	1.651 (41.94) 1.631 (41.43)	1.745 (44.32) 1.725 (43.82)	1.222 (31.04) 1.214 (30.84)	1.235 (31.37) 1.225 (31.12)	1¼-20UN-2A	1.265 (32.13) 1.255 (31.88)

Shell size	AR° BSC	BR° BSC	CR° BSC	DR° BSC	Panel O ring	FE ' BSC Position 1	2	3	4	5	6
8	128°	—	—	—	MS9068-018	40	40	40	40	40	40
10	105°	140°	215°	265°	MS9068-020	60	46	46	60	89	60
14	105°	140°	215°	265°	MS9068-024	60	46	46	60	89	60
16	105°	140°	215°	265°	MS9068-026	50	50	50	50	50	50
18	105°	140°	215°	265°	MS9068-028	50	50	50	50	50	50

INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.090	2.29	.770	19.56
.005	.13	.094	2.39	.778	19.76
.006	.15	.125	3.18	1.275	32.39
.015	.38	.200	5.08	1.340	34.04
.020	.51	.230	5.84	1.305	33.75
.067	1.70	.482	12.24	1.400	35.56
.073	1.85	.502	12.75		

NOTES
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions are after plating.
5. Normal mating keyway position shown For alternate keyway positions see MIL-C-81511, figure 4.
6. Distance from front of shell to the point at which a gage pin, having the same basic diameter as plug shell, engages the spring member
7 For shell sizes 20, 22 and 24 use MIL-C-81511/23 connectors

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REQUIREMENTS

Dimensions and configuration: See figure and table I

Series 2

Classes A, F and E

Shell size See table II

Insert arrangement See supplement for MS sheet corresponding to desired shell size

Contacts: Style P or S, crimp type

Shell polarization 1 thru 6, key position

Weight See table II

Marking Part number on shell

Mating counterpart Plug connectors specified in MIL-C-81511 6 and '34

Insertion and removal tool As specified in MS3323, MS3342 and MS3344

Slotted nut Shall conform to MIL-C-81511/20

Part number The part number shall consist of M81511 03 together with the applicable codes for class, shell size, insert arrangement, contact style, and key positions Information concerning the formulation of the part number can be found in MIL-C-81511

TABLE II

Shell size	Weight (REF) 1/ (pounds)
18 (code F)	090
16 (code E)	078
14 (code D)	063
10 (code B)	040
8 (code A)	030

1/ Removable crimp contacts not included in weight
(See contact drawings for weight of pin and socket contact)

Custodians
Army - EL
Navy - AS
Air Force - 17

Preparing activity
Navy - AS

Review activities:
Army - MI, MU, AV, WC
Navy - SH, EC, OS
Air Force - 11, 80
DSA - ES

(Project No. 5935-1732-3)

User activities:
Army - ME, AT
Navy - MC
Air Force -

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel (either Government or contractor) involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-81511/3E - CONNECTORS, ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISCONNECT, ENVIRONMENT RESISTING; AND ACCESSORIES: RECEPTACLE, JAM NUT MOUNT, CRIMP-TYPE CONTACTS, CLASS A, F AND E (SERIES 2)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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DEPARTMENT OF THE NAVY

Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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NAVY DEPARTMENT

OFFICIAL BUSINESS

Weapons Engineering Standardization Office (Code X)
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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